

The Data-Driven Commute: How Cab Analytics Improve City Mobility

CSP 571 Data Preparation and Analysis

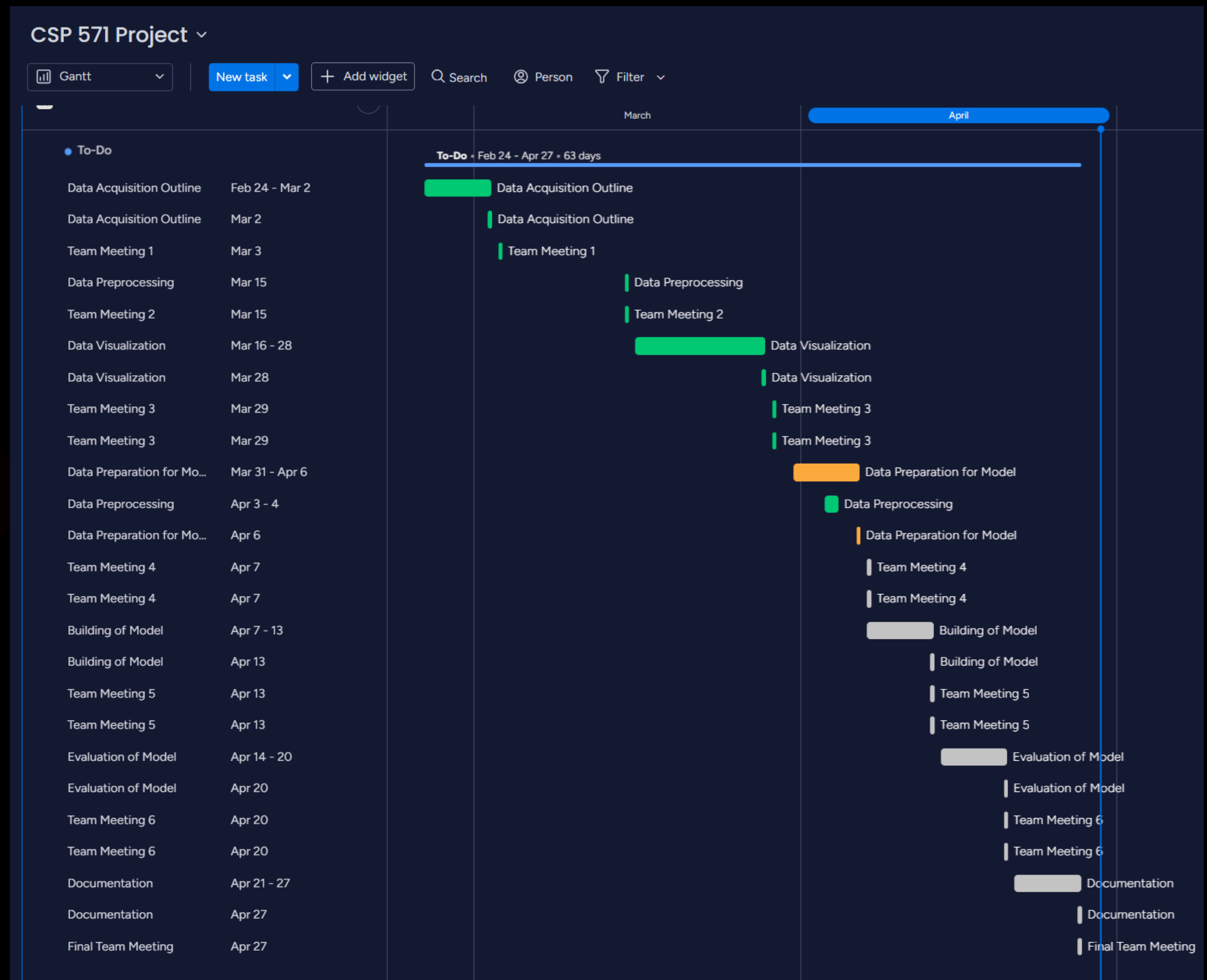
Team Members

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Project Plan and Details



Problem Statement

The project aims to analyze pricing and demand patterns in Uber and Lyft ride-sharing services, focusing on how external factors like weather influence ride costs and availability. Using a large dataset of ride details and environmental conditions, the study will identify key factors affecting prices, compare service patterns between companies, and build predictive models to forecast ride prices. The goal is to uncover insights that can help optimize urban mobility and improve the ride-hailing experience.

Dataset Description

Dataset Link = <https://www.kaggle.com/datasets/jonathanoheix/face-expression-recognition-dataset>

693,071

Rows

57

Columns

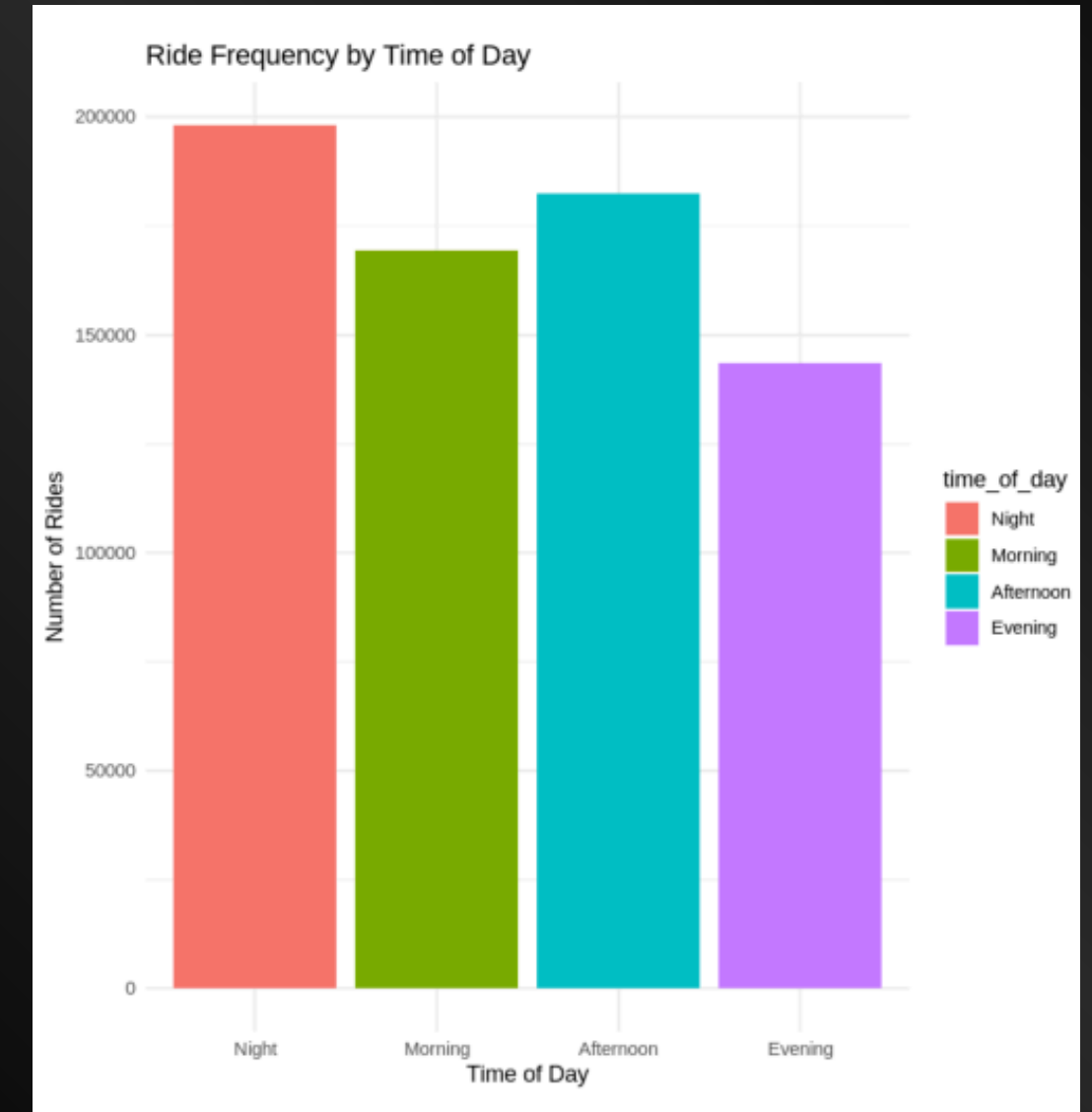
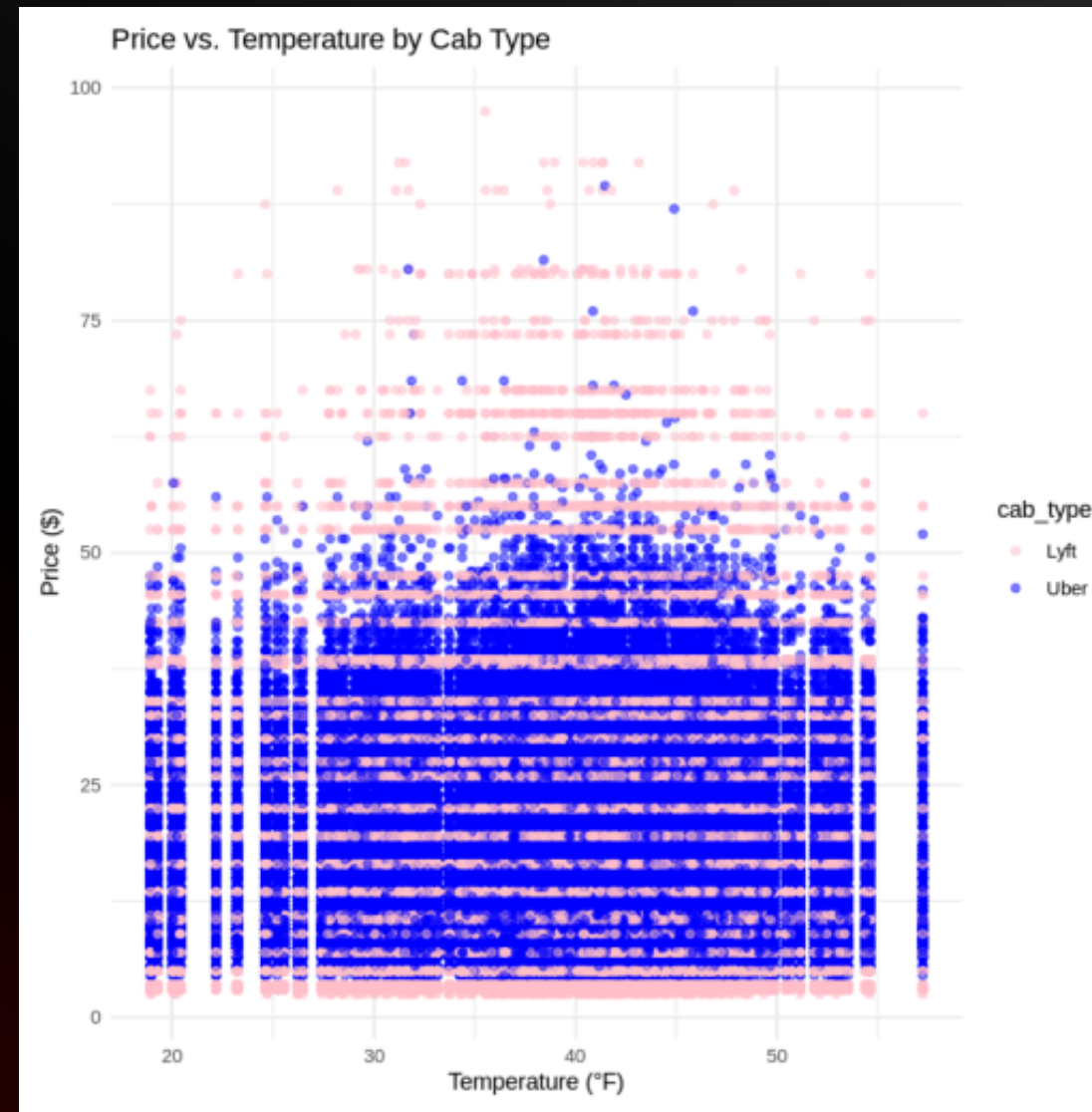
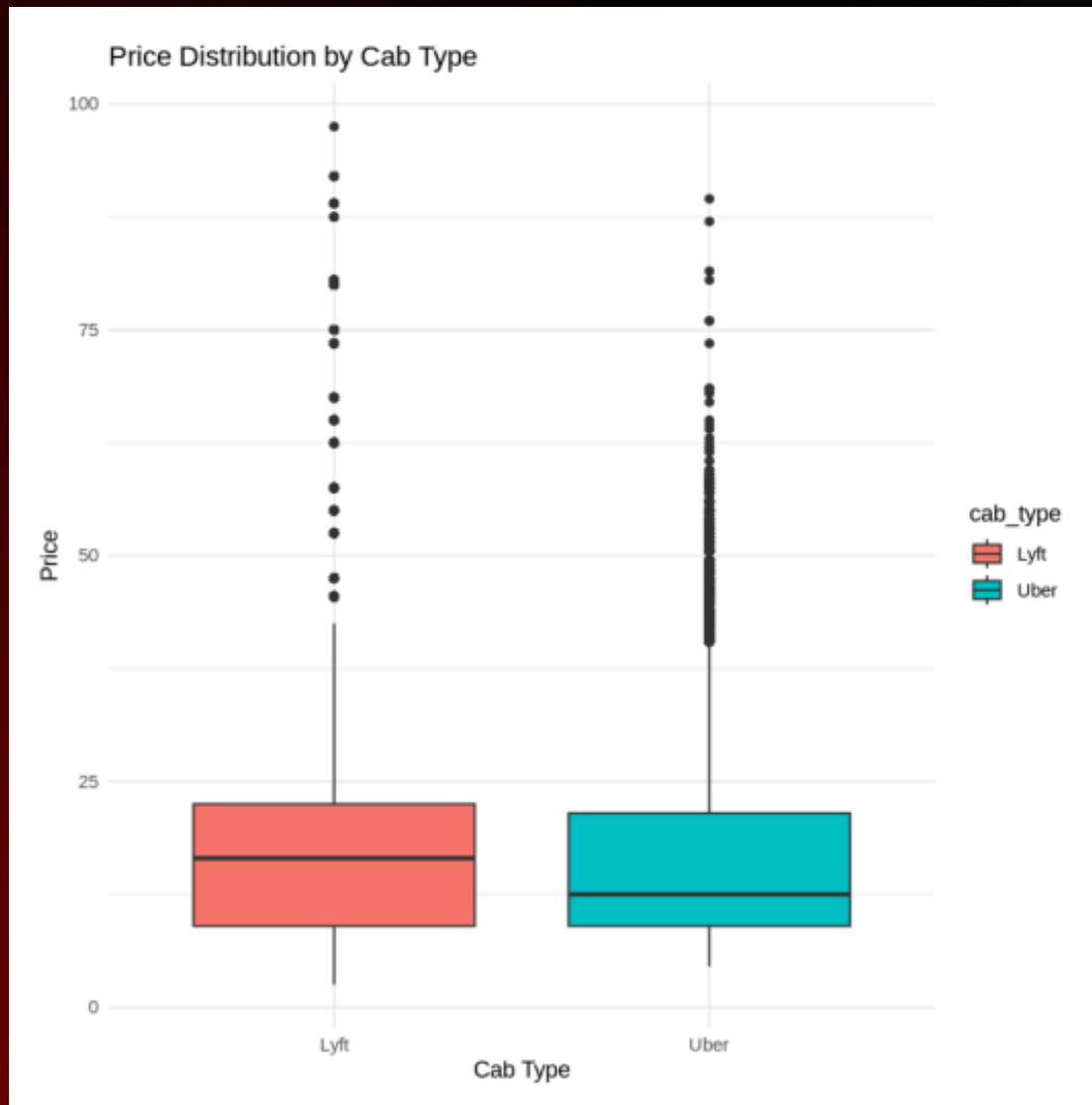
23 Features

Fri	Lyft XL	UberPool	Partly Cloudy
Sat	Lux Black XL	UberXL	WAV
Sun	Lux Black	Black	Possible Drizzle
Shared	surge_multiplier	Black SUV	Overcast
Mostly Cloudy	Drizzle	Rain	Light Rain
distance	Partly Cloudy	Foggy	-

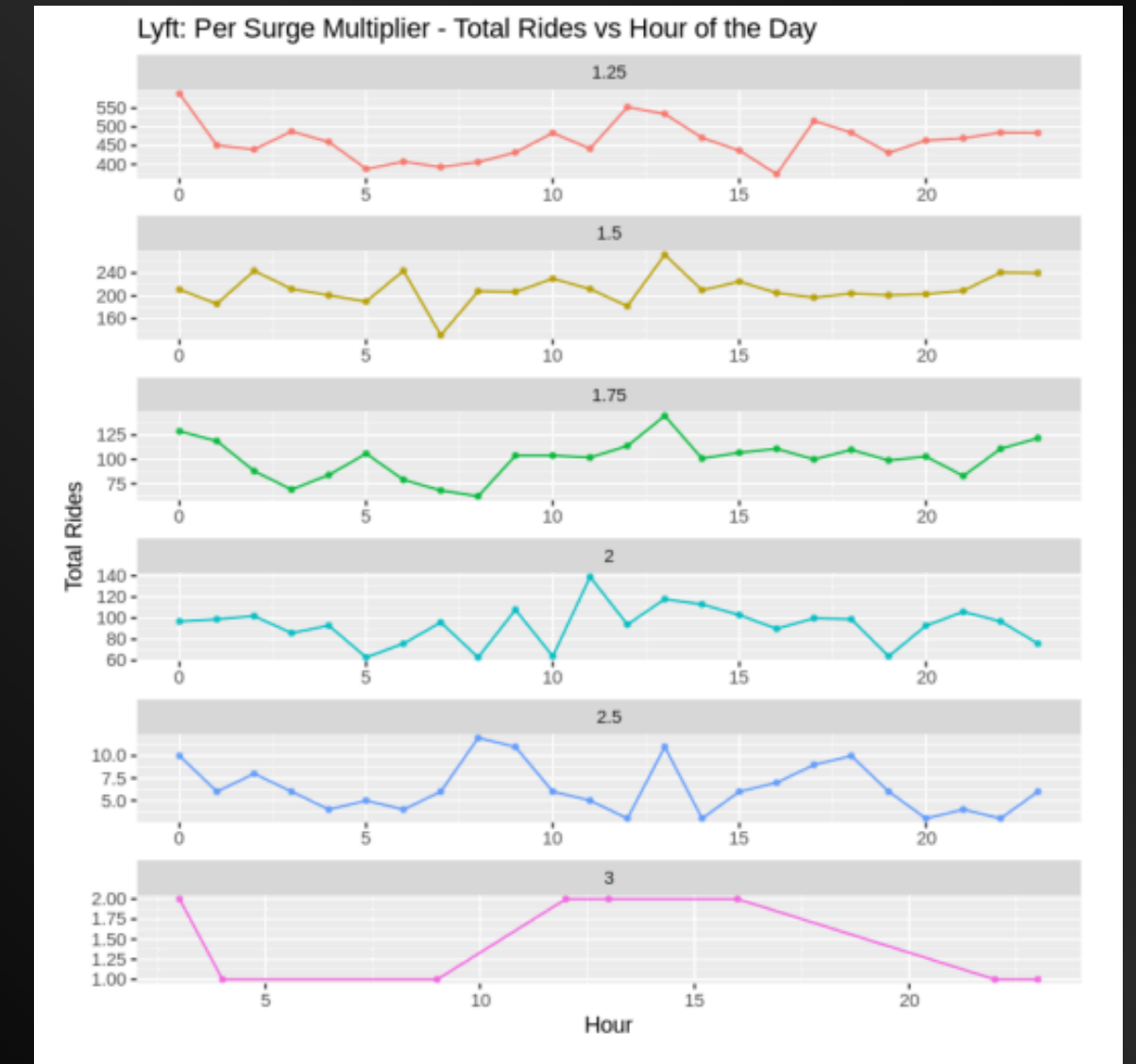
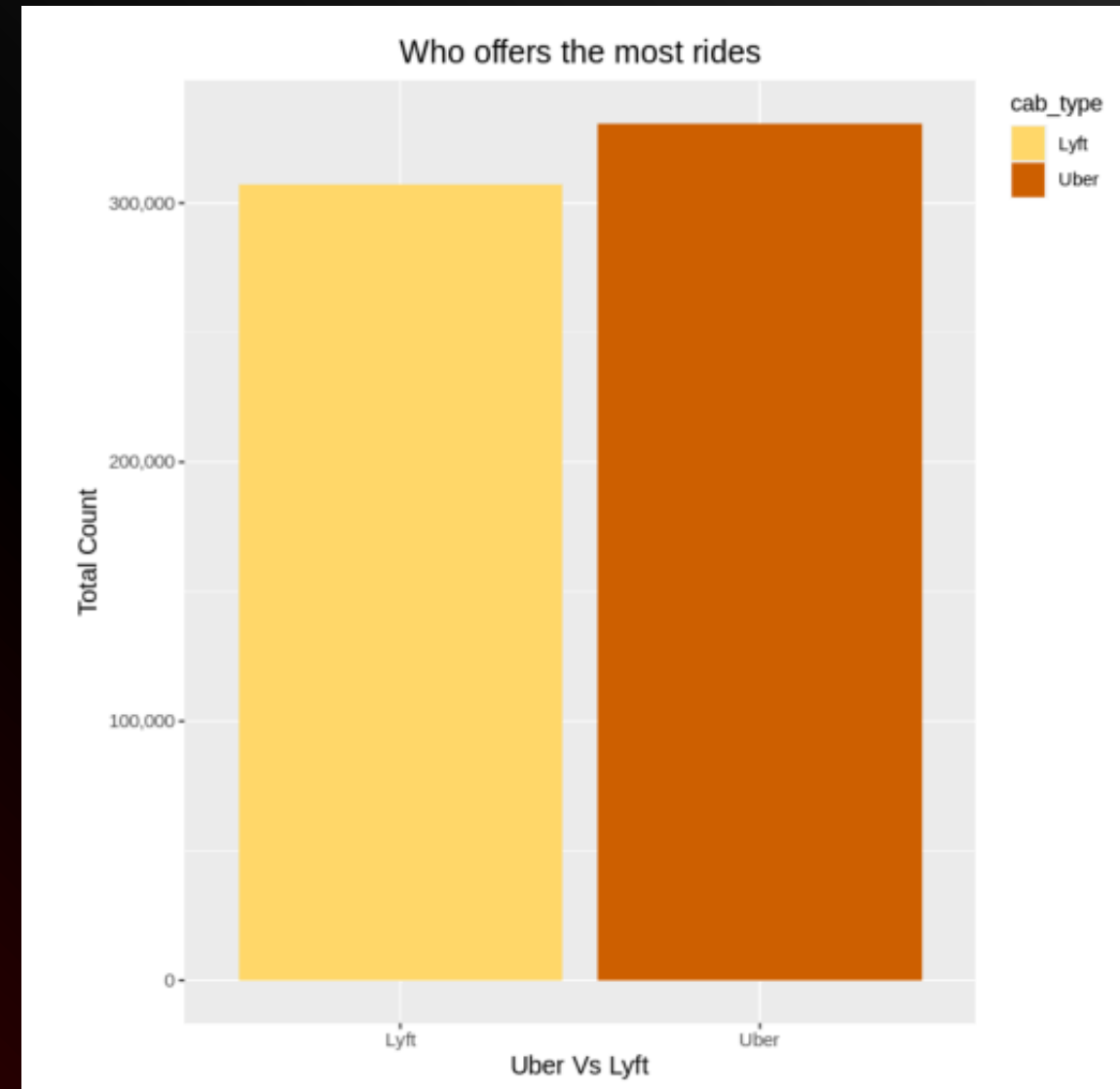
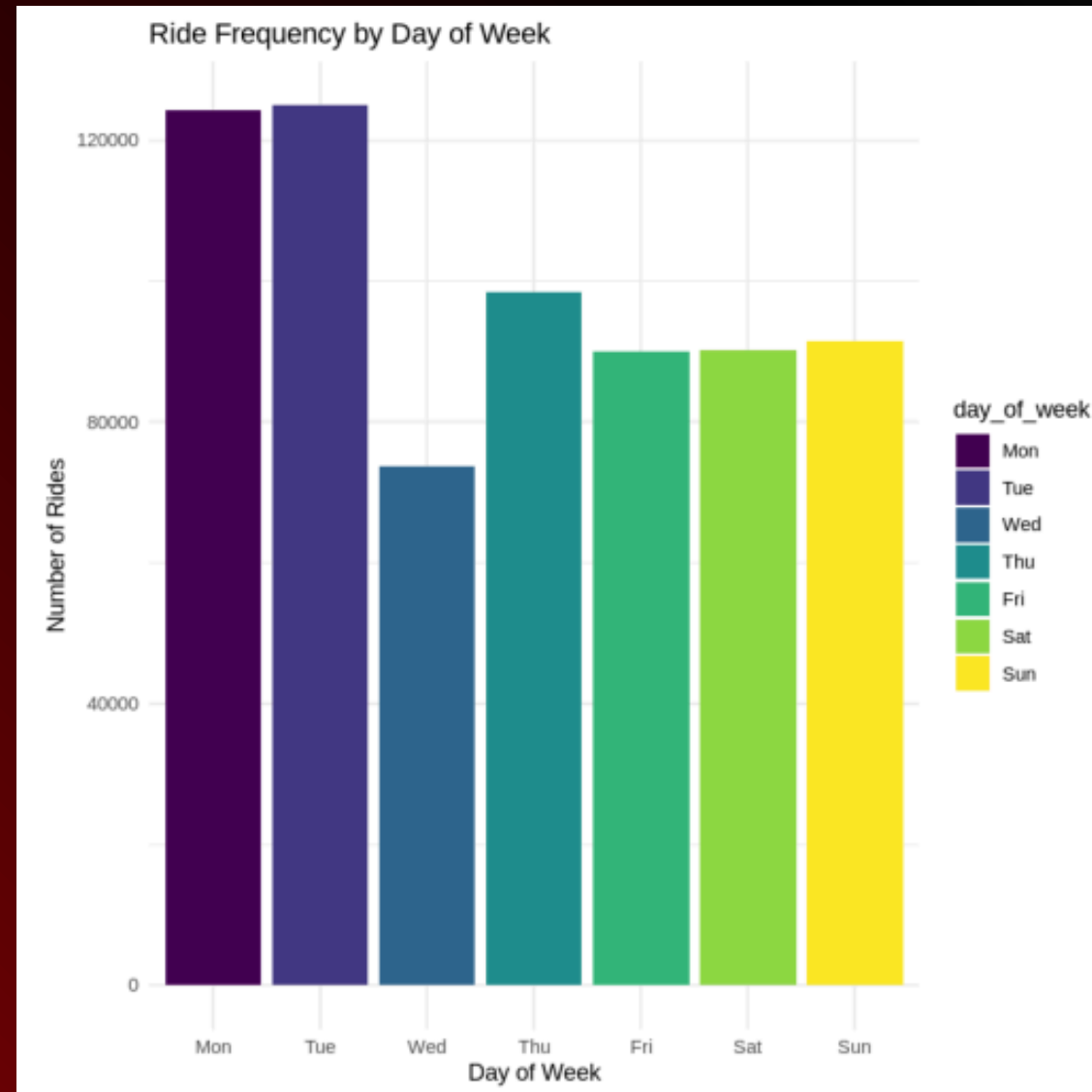
Questions **Answered**:

- Who offers the most rides, Uber or Lyft?
- Total Rides vs Hour of the Day
- Minimum and maximum fare prices
- Top 10 most Popular Stations
- Weather affect on the rides
- Temperature affect on the ride's price
- Trips By Hour and Month
- Price range between Uber and Lyft

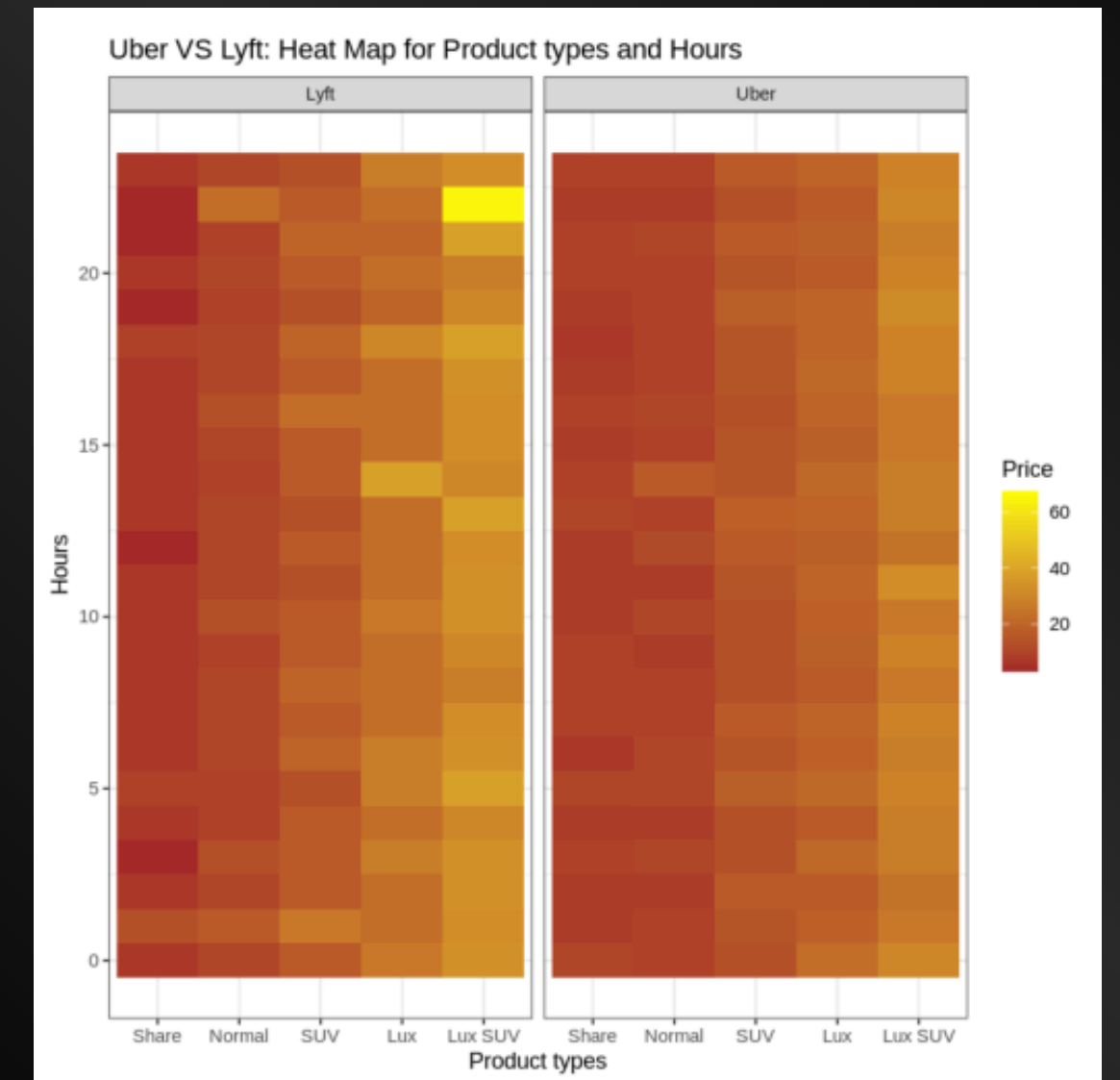
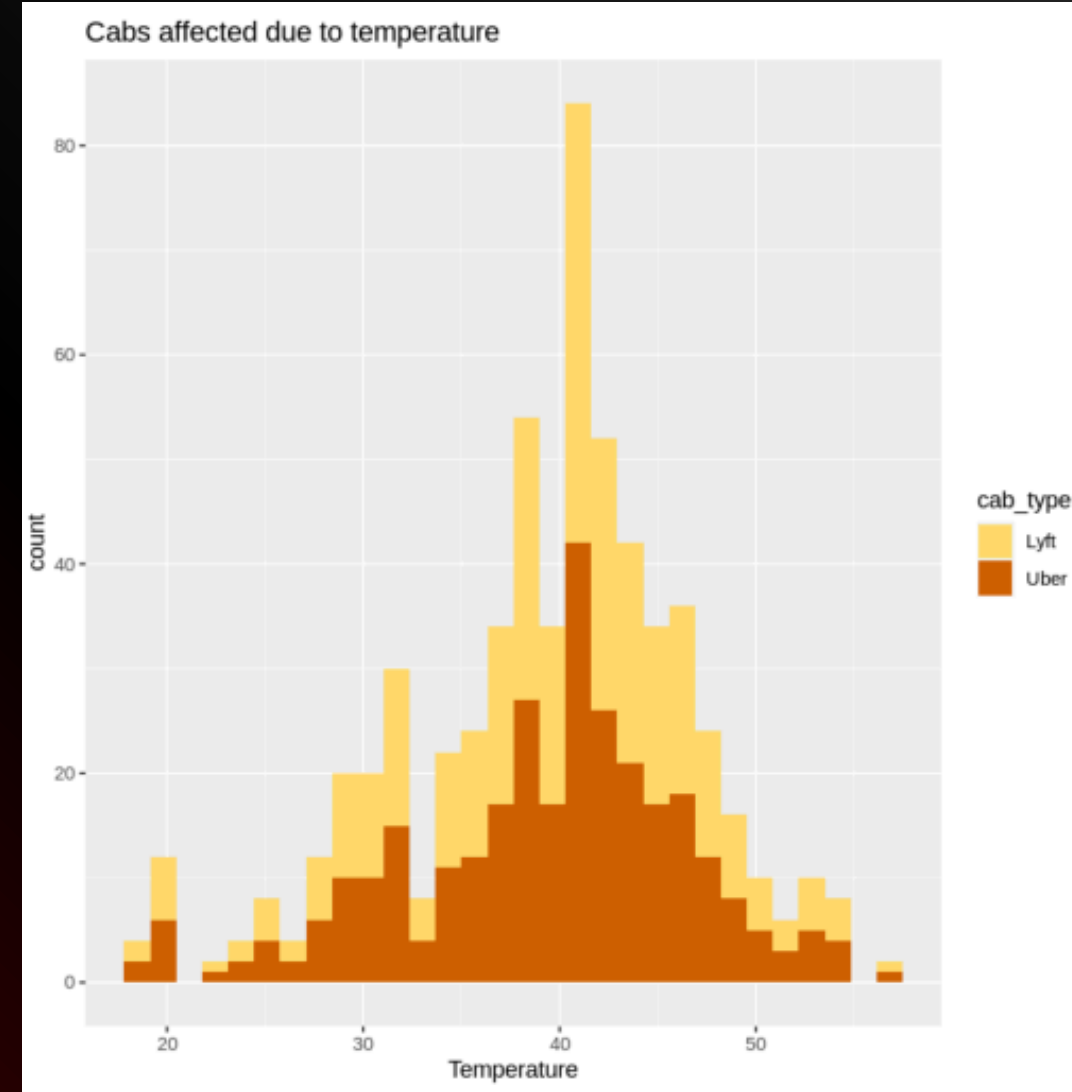
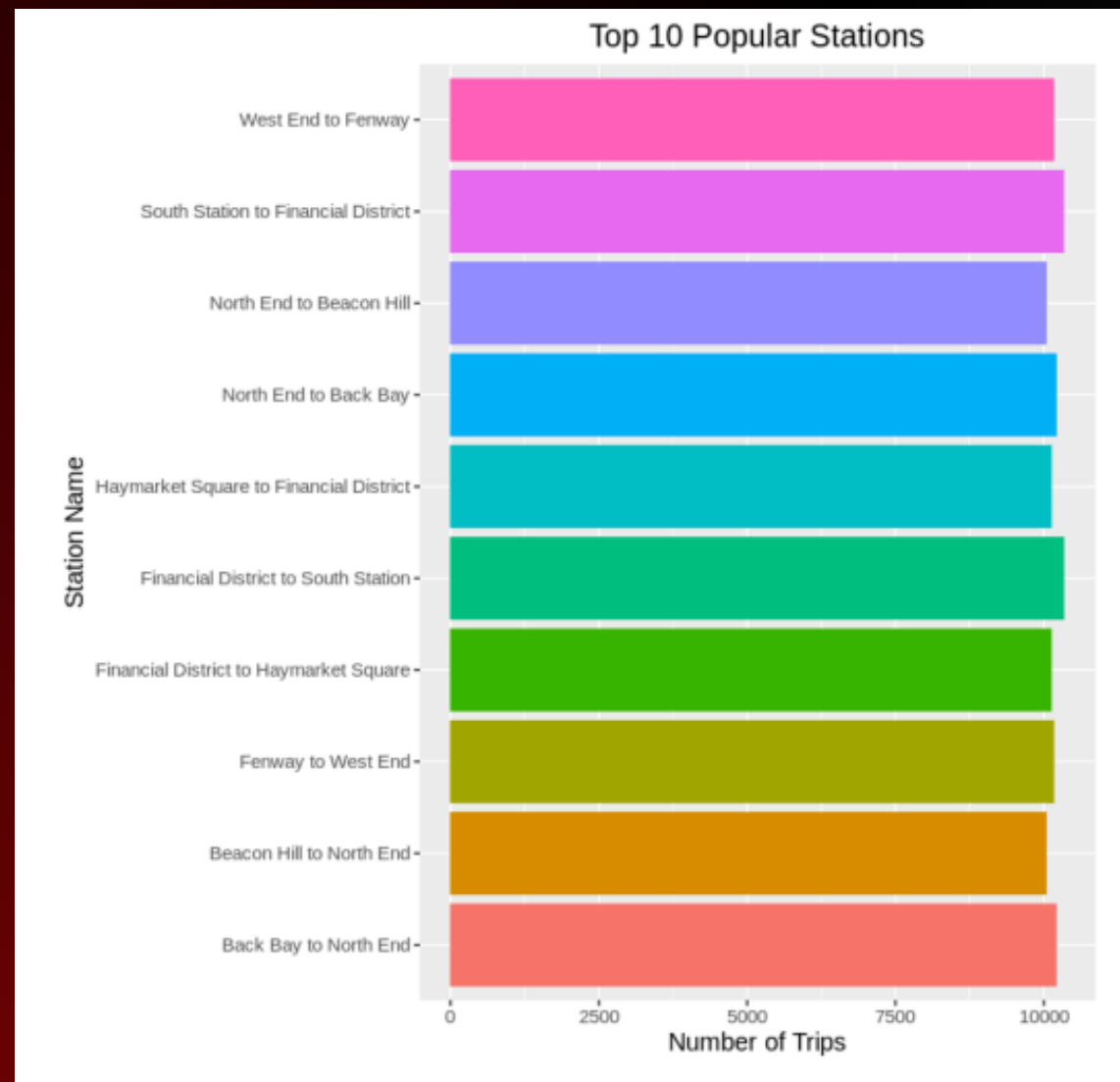
Exploratory Data Analysis



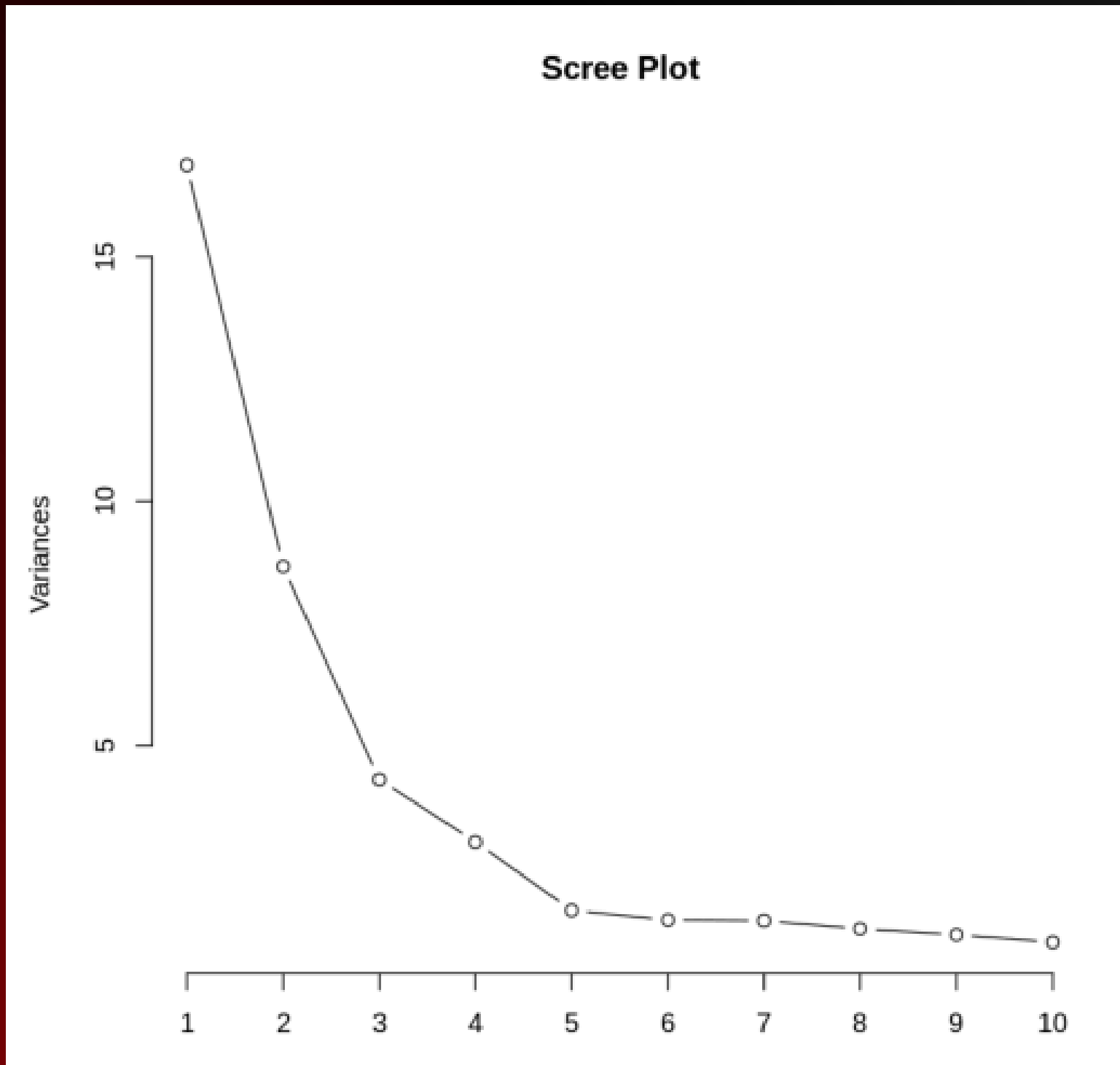
Exploratory Data Analysis



Exploratory Data Analysis



Principle Component **Analysis**



The chart shows that most of the important information sits in the first few factors. The very first factor carries a lot on its own, and the line drops sharply right after it. The second and third factors still add useful detail, but each factor after them contributes only a small amount. From the fourth factor onward, the curve levels off, so keeping more factors offers little value. In short, relying on just the first three factors should capture the key patterns while keeping the analysis simple.

Model Selection

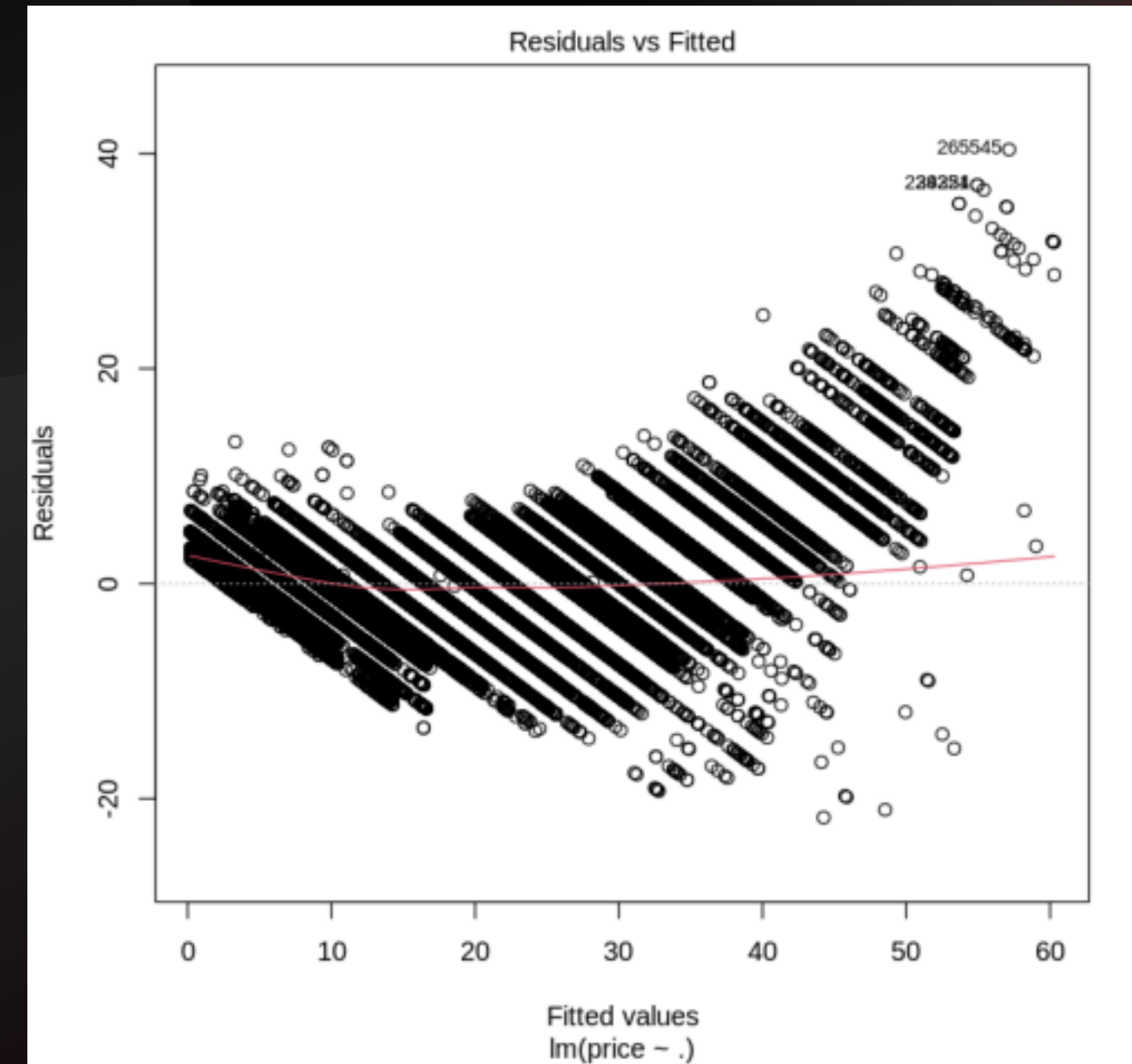
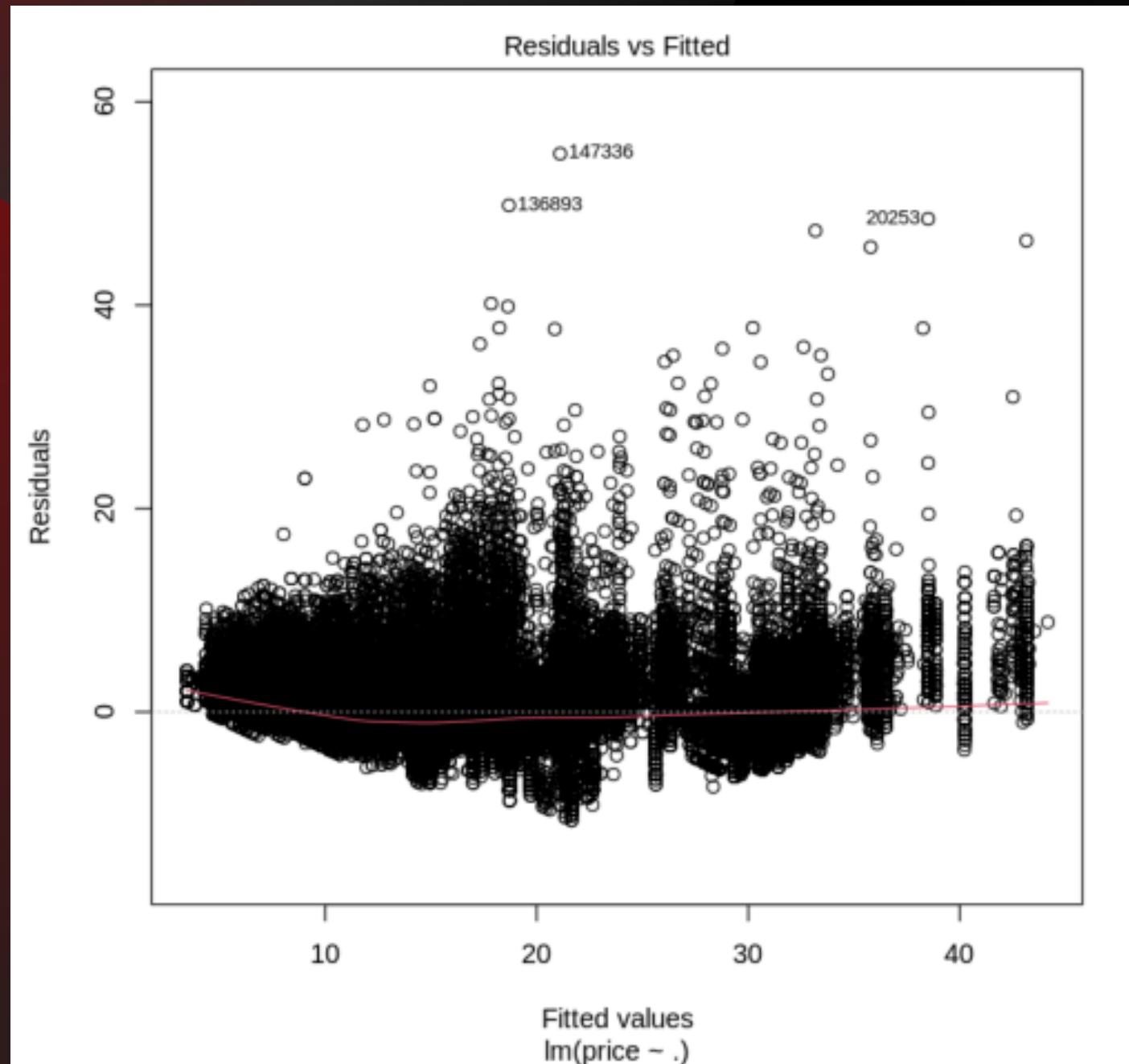
In the pursuit of an optimal predictive model for ride-sharing prices, we have employed three distinct statistical learning methods: Linear Regression, Decision Trees, and Random Forest. The following is a detailed analysis of the model selection process for both Uber and Lyft datasets.

Linear
Regression

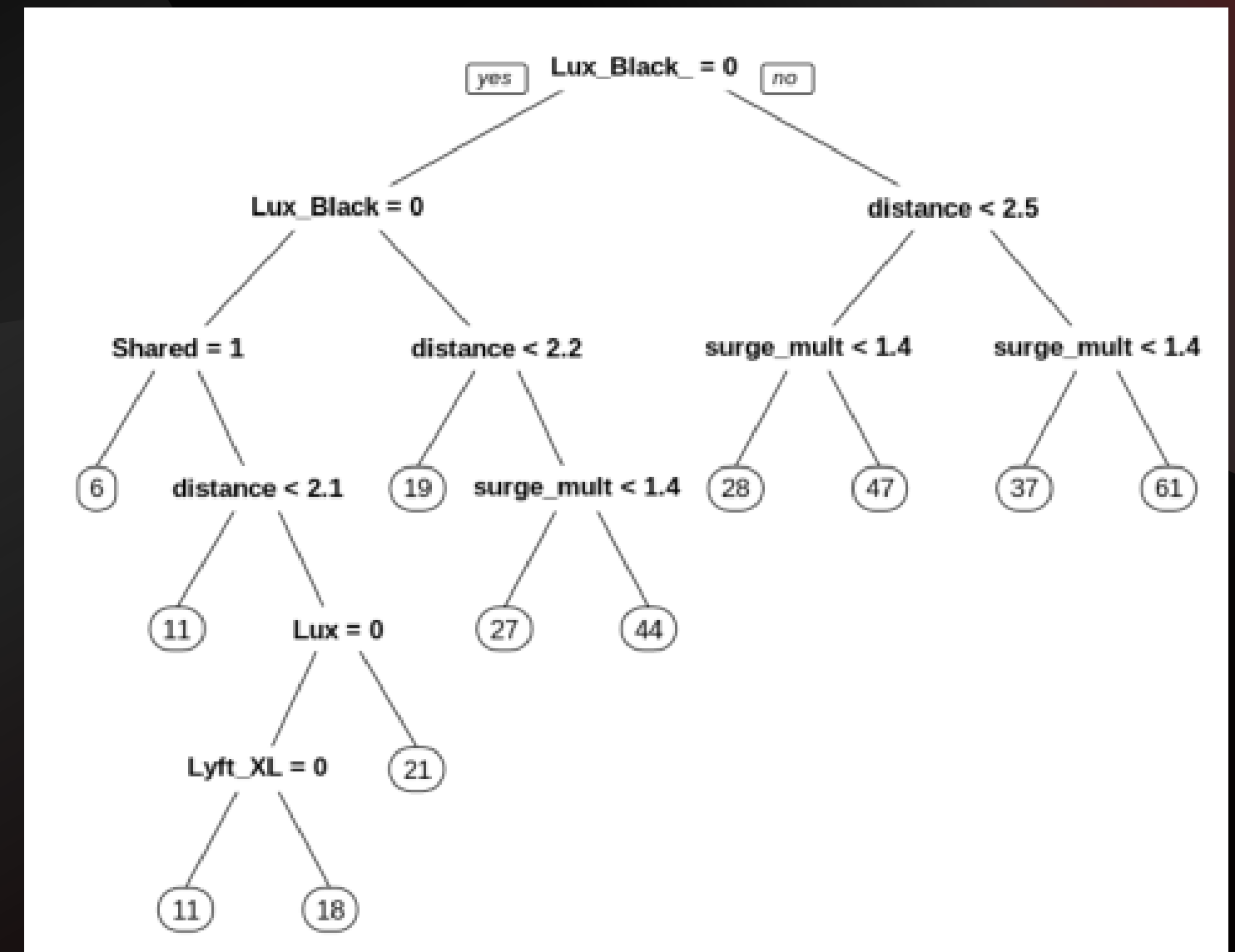
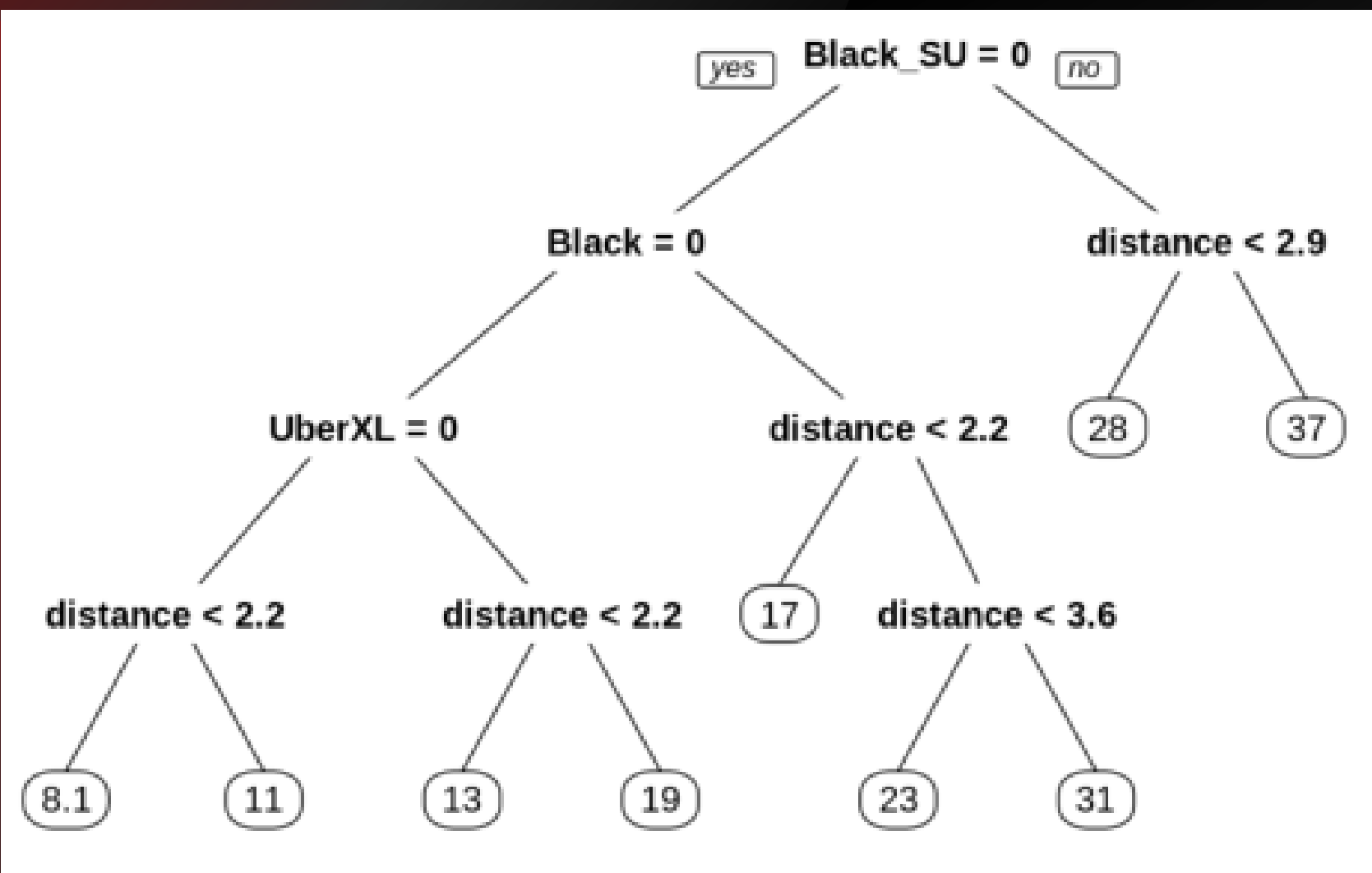
Decision
Tree

Random
Forest

Linear Regression



Decision Tree



Random Forest

```
A matrix: 2 × 2 of type dbl
```

	actuals	predicted
actuals	1.0000000	0.9592433
predicted	0.9592433	1.0000000

```
mae      mse      rmse      mape
1.7313295 6.5205389 2.5535346 0.1290741
```

```
'The Accuracy of Random Forest for Uber :87.092593'
```

Uber

```
A matrix: 2 × 2 of type dbl
```

	actuals	predicted
actuals	1.0000000	0.9775778
predicted	0.9775778	1.0000000

```
mae      mse      rmse      mape
1.666995 5.217596 2.284206 0.128877
```

```
'The Accuracy of Random Forest for Lyft :87.112304'
```

Lyft

Conclusion

- The linear regression model delivers stronger results for Uber than for Lyft on every evaluation measure, implying it captures the link between input variables and fare levels more effectively for Uber. In contrast, Lyft's elevated MSE and RMSE reveal greater price variability that the linear approach fails to model precisely.
- Compared with linear regression, the decision-tree model performs worse; its error metrics are higher and its accuracy lower. This shortfall likely stems from overfitting—the tree memorizes noise in the training data, so its predictions do not generalize well.
- The random-forest model lands in the middle ground, tempering bias and variance and avoiding the overfitting seen in a single decision tree. By averaging the predictions of many trees, it achieves solid overall accuracy and better robustness on new, unseen data.

Future **Work**

- Exploring more sophisticated machine-learning approaches—such as neural networks or gradient-boosting methods—could uncover subtler patterns in the data. It may also prove fruitful to design composite or hybrid models that fuse the advantages of multiple algorithms to boost predictive power.
- A further priority is to roll these models out in real-time prediction pipelines, monitor their behavior in production, and refine them based on live feedback and any performance gaps that surface. Building an adaptive learning setup that refreshes the models as fresh data streams in would help them stay accurate and up-to-date.
- Lastly, dynamic-pricing systems must be examined through an ethical lens to make sure the algorithms do not unintentionally introduce price discrimination or other societal harms. Addressing this concern calls for collaboration across disciplines, blending data-science expertise with perspectives from social sciences and ethics.



Thank you!